

Business Proposal

Continuous Emergency Department Safety Monitoring & Hybrid Decision Support

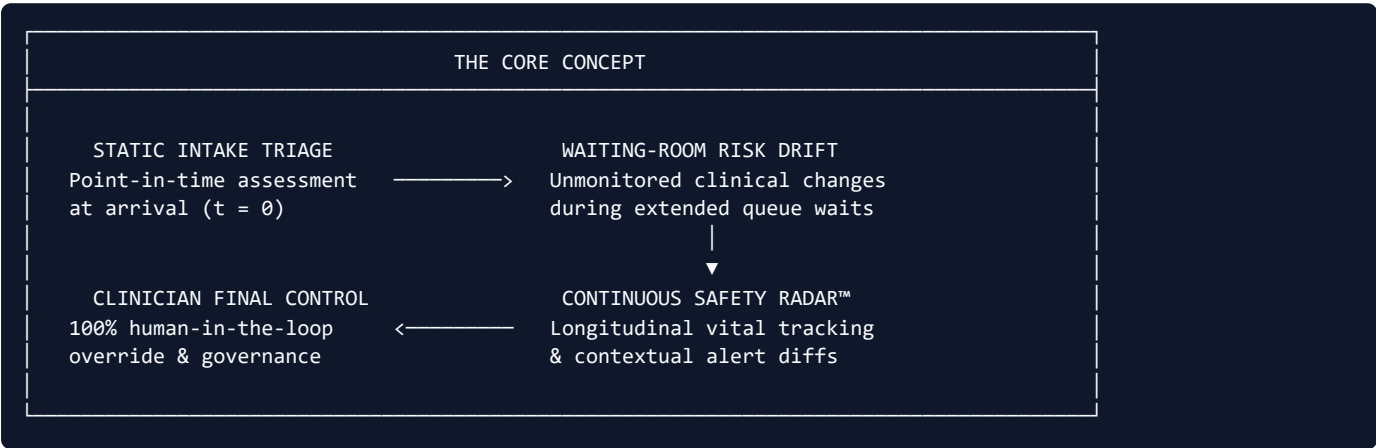
From static triage to continuous waiting-room safety awareness

Document Type: Competition Business Proposal & Executive Strategy

System Status: Functional Decision-Support Prototype

Evaluation Evidence: Synthetic Demonstration Benchmarks & Exploratory Generalization

Repository: https://github.com/sahilsheoran999/patient_triage



PROTOTYPE SIMULATION & REGULATORY NOTICE

PatientTriage.ai is a clinical decision-support prototype designed for emergency department triage surveillance.

- **Prototype Demonstration:** All operational workflows, triage trajectories, and simulation models are demonstrated on synthetic datasets.
- **No Autonomous Clinical Action:** The software does not diagnose medical conditions, recommend drug therapies, or replace human clinical judgment.
- **Hierarchy of Authority:** Licensed healthcare practitioners retain 100% final decision authority. Deterministic safety rules establish non-downgradable safety floors that advisory machine learning models cannot lower.
- **Regulatory Posture:** Commercial deployment is subject to formal retrospective validation, prospective clinical trials, Institutional Review Board (IRB) approval, cybersecurity certification, and regulatory clearance (e.g., SaMD review by CDSCO, US FDA, or European CE mark authorities).

1. Executive Summary & Table of Contents

Emergency departments (EDs) worldwide face unprecedented operational challenges: severe patient overcrowding, acute clinical nursing shortages, and unpredictable volume surges. Under conventional five-level triage systems (e.g., ESI, MTS), patients receive a static, one-time acuity rating at intake. However, human physiology is dynamic; during prolonged waiting periods stretching into hours, patients initially categorized as moderate or low acuity can silently deteriorate before receiving bedside evaluation.

PatientTriage.ai transforms emergency triage from an isolated intake snapshot into an active, continuous safety monitoring workflow by pairing deterministic physiological safety rules with advisory machine learning under strict clinician authority.

Four Core Strategic Pillars

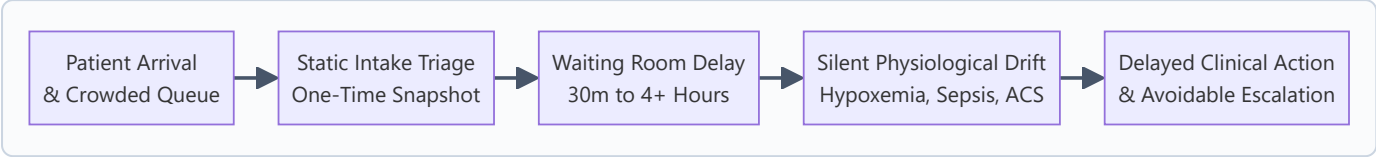
 CONTINUOUS SURVEILLANCE Waiting-Room Radar™ tracks longitudinal vital deltas and wait times in real time.	 HYBRID SAFETY FUSION Deterministic rules enforce non-downgradable safety floors over ML predictions.
 EXPLAINABLE AI AT POINT Feature-level SHAP values display mathematical drivers behind advisory outputs.	 CLINICIAN FINAL AUTHORITY 100% human-in-the-loop governance with mandatory reason codes and audit logs.

Document Table of Contents

- 1. Executive Summary & Table of Contents — Page 2
- 2. Problem Framing & The Waiting-Room Safety Gap — Page 3
- 3. Solution Overview & 5-Component Architecture — Page 4
- 4. Why the Hybrid Model: Rules + ML + Clinician Authority — Page 5
- 5. Target Users & Stakeholder Value Ecosystem — Page 6
- 6. Product Experience & Clinical Workflow — Page 7
- 7. Business Value Pillars & Pilot Value Hypotheses — Page 8
- 8. Competitive Differentiation Matrix — Page 9
- 9. Responsible AI & Clinical Governance Guardrails — Page 10
- 10. Phased Deployment Strategy & Validation Gates — Page 11
- 11. Phased Product Roadmap — Page 12
- 12. Comprehensive Risk & Mitigation Matrix — Page 13
- 13. Scalability Architecture & Commercialization Models — Page 14
- 14. Prototype Evidence & Hospital Integration Feasibility — Page 15
- 15. Conclusion & Strategic Call to Action — Page 16

2. Problem Framing & The Waiting-Room Safety Gap

Emergency departments face a fundamental structural dilemma: triage assessments occur at minute zero ($t = 0$), yet patients spend substantial time waiting in crowded waiting rooms where clinical conditions can evolve unnoticed.



The Safety Gap: Initial Risk ≠ Permanent Risk

$$\text{Patient Risk}(t) = f(\text{Vitals}(t), \text{Symptoms}(t), \text{Elapsed Wait Time}(t - t_0), \Delta\text{Physiology}(t))$$

Conventional triage systems treat $\Delta\text{Physiology}(t) = 0$, creating a dangerous operational gap where clinical deterioration is missed between registration and physician examination.

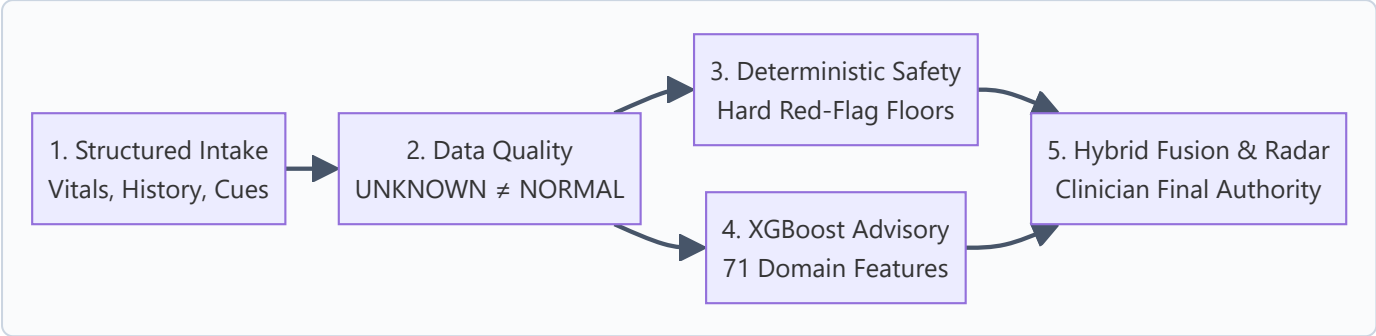
THREE WAITING-ROOM TRAJECTORIES		
PATIENT TYPE	INTAKE PRESENTATION	WAITING-ROOM DRIFT & RADAR INTERVENTION
1. Stable Patient	SpO ₂ 98%, HR 74, Temp 36.8°C Minor wrist pain (ESI-4)	Vitals remain stable across 45 minutes. Radar maintains `SAFE` queue state.
2. Deterio-rating	SpO ₂ 95%, HR 86, Mild cough Moderate baseline risk	At t=40m, SpO ₂ drops to 87% ($\Delta = -8\%$). Radar triggers immediate `ESCALATE` diff.
3. Incomplete Data	Chest discomfort, Levine sign Unmeasured BP (null/missing)	Missing BP elevates uncertainty score. Radar flags `WATCH` + Prompts re-check.

Core Clinical Drivers of Waiting-Room Risk

- **Silent Decompensation:** Conditions like occult sepsis and atypical acute coronary syndromes progress without immediate overt respiratory distress.
- **Cognitive Saturation:** During volume surges, triage staff managing 50+ patients cannot manually recalculate trend lines or monitor time limits.
- **Unsafe Imputation:** Rushed staff frequently assume omitted vitals are normal, concealing high-risk presentations.

3. Solution Overview & 5-Component Architecture

PatientTriage.ai replaces static triage scoring with an end-to-end clinical decision-support ecosystem built on five core architectural components:



FIVE CORE ARCHITECTURAL COMPONENTS	
1. Patient Intake	Rapid, structured capture of vital signs, chief complaints with negation protection, past medical history, and clinical signs.
2. Data Quality	`UNKNOWN ≠ NORMAL` governance: unmeasured vitals reduce data completeness and elevate uncertainty rather than defaulting.
3. Deterministic Safety Rules	Non-negotiable physiological red-flag thresholds that establish unbreachable `CRITICAL` safety floors (Safety Authority).
4. XGBoost Advisory Intelligence	71-feature multi-class ensemble providing continuous probability estimates and SHAP mathematical attributions (Advisory Signal).
5. Continuous Radar & Human Control	Waiting-Room Radar™ longitudinal surveillance tracking elapsed wait times and vital deltas under 100% Clinician Authority.

Clear Authority Separation

- CLINICIAN: Final Decision Authority (100% Override Authority with Audit Trail)
- SAFETY RULES: Safety Authority (Unbreachable CRITICAL Physiological Floors)
- XGBOOST MODEL: Advisory Signal Only (71 Features; Cannot Downgrade Safety Floors)
- DATA QUALITY: Safety Constraint (`UNKNOWN ≠ NORMAL` Uncertainty Enforcement)

4. Why the Hybrid Model: Rules + ML + Clinician Authority

A fundamental product differentiator of PatientTriage.ai is its rejection of both "pure rules" and "pure machine learning" in favor of a **Hybrid Safety Fusion** paradigm:

THREE-WAY PARADIGM COMPARISON		
PARADIGM	CORE CAPABILITIES	SYSTEMIC LIMITATIONS
Pure Rules Engine	- Deterministic & auditable - Verifiable safety bounds 100% predictable outcomes	- Rigid boundary thresholds - Blind to non-linear patterns - High false-negative rate (45%)
Pure Machine Learning	- Complex multi-factor risk - High sensitivity (98.9%) - Continuous probabilities	"Black box" unpredictability - Unsafe edge-case downgrades - Vulnerable to missing data
PATIENTTRIAGE.AI HYBRID SAFETY FUSION	- Best of both paradigms - Absolute safety floors - Advanced pattern discovery - Explainable SHAP factors	- Requires dual-rule maintenance - Dependent on clinician review - Prototype validation needed

Key Architectural Strengths

WHY HYBRID SAFETY FUSION WINS	
1. Safety Floor Enforcement	When a patient presents with extreme vitals (e.g., SpO ₂ < 88%), deterministic rules force CRITICAL. ML cannot downgrade acuity.
2. Subtle Risk Detection	In complex multi-symptom presentations where individual vitals are borderline, XGBoost detects interactions and upgrades acuity
3. Clinician Trust & Adoption	Clinicians trust systems that have explicit, verifiable guard-rails rather than opaque probability distributions alone.

Central Governance Doctrine:
"Machine learning advises. Deterministic safety rules protect. Clinicians decide."

5. Target Users & Stakeholder Value Ecosystem

PatientTriage.ai creates aligned clinical and operational value across all acute-care stakeholders:

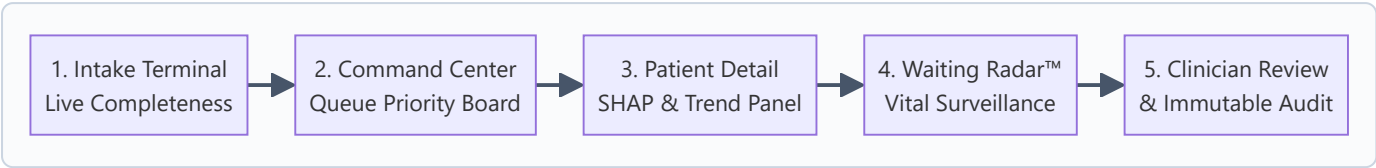
STAKEHOLDER VALUE MATRIX		
STAKEHOLDER	CORE OPERATIONAL PAIN POINT	PATIENTTRIAGE.AI VALUE DELIVERED
 Triage Registered Nurses (RNs)	• Cognitive surge overload • Unmonitored waiting queues • Fear of AI black boxes	• Automated longitudinal radar • Contextual "Why Now?" alerts • 100% human override authority
 ED Leadership & Flow Coordinators	• Bottlenecks & wait times • Unpredictable surge crises • Regulatory compliance	• Surge Mode volume management • Real-time queue visibility • Standardized reassessment time
 Clinical Safety & Risk Management	• Diagnostic drift liability • Unexplained AI decisions • Missing data vulnerability	• Tamper-evident audit logging • Feature-level SHAP evidence • Zero silent default imputation
 Patients & Families	• Long anxious wait times • Risk of silent in-queue decompensation	• Timely clinical escalation • Safer waiting-room environment

Four Value Pillars Created

 SAFETY AWARENESS Real-time vital-trend delta surveillance in waiting room.	 WORKFLOW VISIBILITY Centralized queue telemetry with Surge Mode adaptation.
 MATHEMATICAL EXPLAINABILITY Local SHAP attributions for every AI advisory output.	 AUDITABLE GOVERNANCE Immutable event stream with mandatory override reasons.

6. Product Experience & Clinical Workflow

The PatientTriage.ai interface is designed for high-stress, low-latency emergency environments:



Five Core User Interface Modules

FIVE CORE INTERFACE MODULES	
1. Command Center	Real-time priority queue featuring risk scores, model class probabilities, uncertainty badges, and Surge Mode controls.
2. Waiting-Room Radar	Longitudinal monitoring grid tracking elapsed wait times against acuity limits with 4 states: SAFE, WATCH, REASSESS, ESCALATE.
3. Patient Intake	Structured triage terminal enforcing explicit null handling (UNKNOWN ≠ NORMAL) with live data completeness gauge.
4. Patient Detail Modal	4-Stat clinical bar, vital trend charts, SHAP feature importance decomposition, and transparent rule factor breakdown.
5. Override & Audit Trail Log	Single-click priority override with mandatory reason codes, clinician notes, and tamper-evident event streaming.

Operational Workflow Efficiency

- **Sub-Minute Intake:** Triage nurses enter structured vitals and clinical cues in < 45 seconds.
- **Zero-Latency Scoring:** Client-side TypeScript inference computes hybrid recommendations in milliseconds.
- **Contextual Alerts:** When deterioration occurs, Radar displays exact vital deltas (ΔSpO_2 , ΔHR) in a "Why Now?" banner, eliminating guesswork.

7. Business Value Pillars & Pilot Value Hypotheses

All economic and operational impacts represent **business hypotheses for future pilot validation** rather than guaranteed claims. Real-world financial impact must be validated through prospective hospital trials.

Four Business Value Pillars

1. PATIENT SAFETY AWARENESS Active surveillance surfaces changing risk continuously.	2. CLINICAL WORKFLOW GAINS Prioritizes nurse attention toward high-risk re-checks.
3. OPERATIONAL VISIBILITY Centralized dashboard tracks surge capacity in real time.	4. GOVERNANCE & AUDITABILITY Complete audit trail of all overrides and recommendations

Potential Operational & Economic Value Hypotheses

- **Earlier Deterioration Capture:** Designed to detect vital desaturations 15–30 minutes earlier during waiting-room delays.
- **Reduced Escalation Severity:** Timely identification of decompensating patients may prevent avoidable ICU transfers from waiting rooms.
- **Nursing Cognitive Relief:** Eliminates manual mental calculations of age-adjusted vitals and wait-time limits during surges.

Proposed Controlled Pilot Success Metrics

PILOT SUCCESS MEASUREMENT PLAN		
PILOT DIMENSION	TARGET SUCCESS METRIC	MEASUREMENT METHODOLOGY
Alert Precision	≥ 80% clinical relevance	Clinician rating on "Why Now?" alerts
Alert Burden	≤ 2.5 alerts/nurse/hour	System telemetry event logging
Response Latency	< 5 minutes to review	Timestamped audit log verification
Deterioration Capt.	0 missed red-flag events	Retrospective safety committee audit
Clinician Agreement	≥ 85% voluntary adoption	Override frequency tracking in audit
Data Completeness	≥ 90% vitals capture rate	Live intake completeness telemetry

8. Competitive Differentiation Matrix

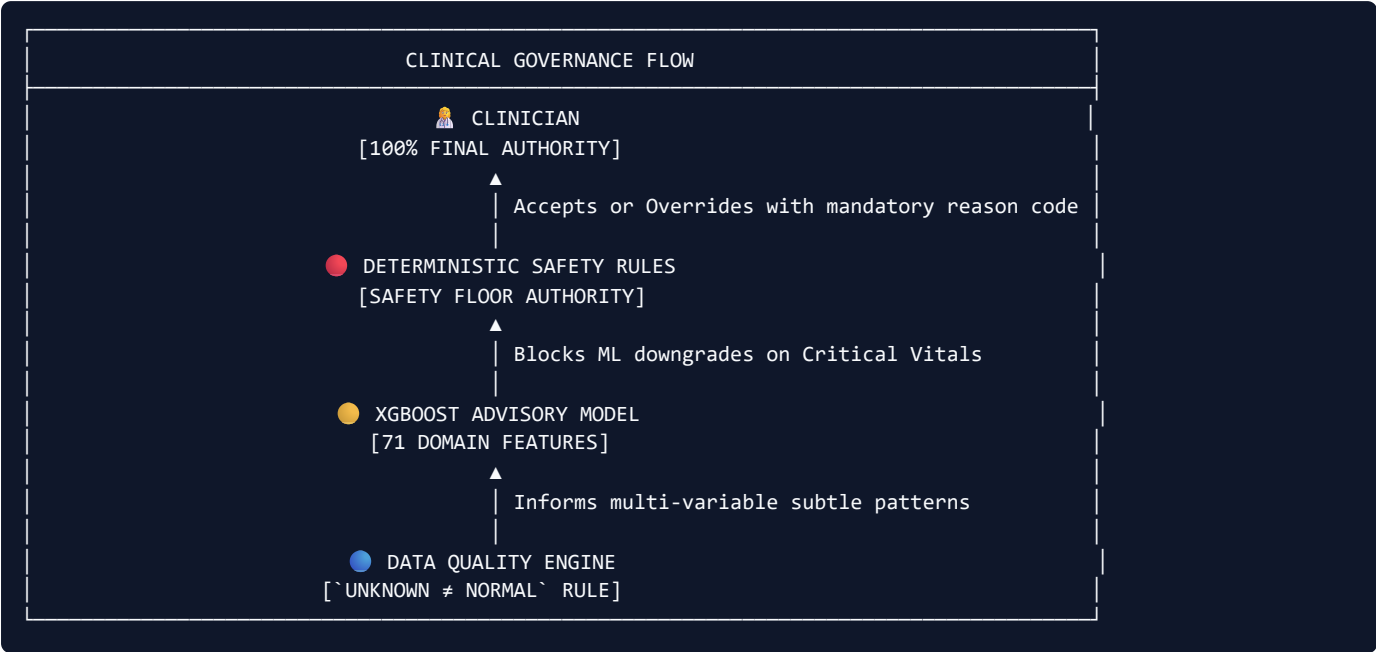
COMPETITIVE LANDSCAPE MATRIX				
CAPABILITY	STATIC ESI / MTS	RULE-ONLY CDSS	BLACK-BOX ML TRIAGE APPS	PATIENTTRIAGE.AI HYBRID PLATFORM
Continuous Queue Radar	✗ No	✗ No	✗ Rare	● **Active Radar™**
Deterministic Safety Flr	⚠ Heuristic	● Yes	✗ No	● **Hard Floors**
Multi-Variable ML Model	✗ No	✗ No	● Yes	● **71 Features**
Transparent SHAP Engine	✗ No	⚠ Limited	✗ Black Box	● **Feature SHAP**
`UNKNOWN ≠ NORMAL` Rule	✗ No	⚠ Incomplete	✗ Imputes	● **Strict Rule**
In-Browser Edge Compute	✗ N/A	⚠ Server	⚠ Heavy Cloud	● **Zero-Latency**
Immutable Audit Logging	⚠ Basic	⚠ Variable	⚠ Variable	● **Event Trail**
Clinician 100% Override	● Manual	● Manual	⚠ Advisory	● **Mandatory Codes**

Strategic Market Positioning

- **Not an EHR Replacement:** PatientTriage.ai is designed as a lightweight decision-support layer that integrates with existing hospital EHR systems via standard FHIR protocols.
- **Safety-First Architecture:** Unlike black-box ML tools that attempt autonomous decision-making, PatientTriage.ai subordinates statistical models to deterministic safety rules and licensed human oversight.
- **Low IT Burden:** In-browser TypeScript execution eliminates expensive server-side GPU hosting requirements.

9. Responsible AI & Clinical Governance Guardrails

PatientTriage.ai is architected strictly in accordance with international Responsible AI and Software as a Medical Device (SaMD) principles:

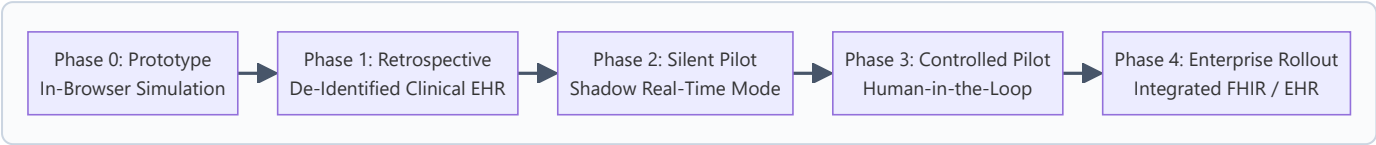


Four Governance Guardrails

FOUR ETHICAL & CLINICAL GUARDRAILS	
1. Clinician Final Authority	AI never replaces human judgement; clinicians retain complete authority to override system recommendations with audit logging.
2. Explainability Boundaries	UI explicitly notes that SHAP values reflect mathematical tree contributions, not pathophysiological causation.
3. Anti-Automation Complacency	Prominent visual disagreement badges require active human confirmation to prevent passive rubber-stamping of AI outputs.
4. Data Minimization & Privacy	System operates strictly on de-identified tokens (<code>`P-***`</code>), maintaining HIPAA and GDPR confidentiality standards.

10. Phased Deployment Strategy & Validation Gates

A responsible, multi-stage deployment model ensures clinical safety, operational readiness, and regulatory compliance before any hospital integration:



FIVE-PHASE DEPLOYMENT GATING		
PHASE	OPERATIONAL FOCUS	MANDATORY EXIT / GATING CRITERIA
Phase 0	In-Browser Simulation (Current State)	- Zero synthetic holdout critical misses 100% functional test passing
Phase 1	Retrospective Validation	- External generalization on multi-center EHR - Statistical calibration curves verified
Phase 2	Silent Shadow Pilot	- Runs in background without nurse visibility - Measure alert frequency & false-positive rate
Phase 3	Human-in-the-Loop Pilot	- Clinician-facing recommendations in 1-2 EDs - IRB oversight & structured nurse feedback
Phase 4	Enterprise Rollout	- Full HL7/FHIR bidirectional EHR integration - Formal SaMD regulatory certification

Gating Policy: Progression to each subsequent phase requires explicit sign-off from clinical safety committees and Institutional Review Boards.

11. Phased Product Roadmap

MULTI-YEAR PRODUCT ROADMAP
PHASE 1: CORE ENGINE & SIMULATION (COMPLETED PROTOTYPE) • 71-feature in-browser XGBoost V2 inference engine • Deterministic safety engine with age-aware vital baseline normalizer • Waiting-Room Radar™ continuous surveillance simulation • Clinician override interface with immutable audit trail
PHASE 2: CLINICAL VALIDATION & CALIBRATION (6-12 MONTHS) • Multi-center de-identified clinical EHR training & validation cohorts • Platt scaling & isotonic probability calibration curves • Subgroup demographic fairness & bias audits across age/gender cohorts
PHASE 3: EHR INTEROPERABILITY & PILOTS (12-18 MONTHS) • HL7 FHIR v4.0 bidirectional connectors (Epic App Orchard, Cerner Marketplace) • Real-time bedside monitor streaming integration (SpO₂, HR telemetry) • Shadow nurse clinical trial in academic medical center emergency department
PHASE 4: SPECIALTY MODULES & REGULATORY CERTIFICATION (18-30 MONTHS) • Pediatric-specialized emergency triage models & physiological bands • Pre-hospital EMS telemetry integration for incoming ambulances • Formal SaMD 510(k) / CE mark regulatory review submissions

Strategic Milestone Focus

- **Near-Term:** Focus entirely on statistical calibration and de-identified clinical EHR generalization.
- **Medium-Term:** Implement standard FHIR connectors to enable zero-friction hospital IT integration.
- **Long-Term:** Expand into pre-hospital telemetry and pediatric emergency pathways.

12. Comprehensive Risk & Mitigation Matrix

RISK MANAGEMENT MATRIX			
IDENTIFIED RISK	IMPACT	ROOT CAUSE	MITIGATION STRATEGY
1. Synthetic Data Generalization	HIGH	Model trained on synthetic distributions	Formal multi-center validation on de-identified clinical EHR
2. Alert Fatigue	MEDIUM	High frequency of radar notifications	Contextual "Why Now?" gating; tunable alert sensitivity
3. ML False Negatives	HIGH	Statistical misestimate on atypical complaint	Deterministic safety floor strictly prevents downgrades
4. Automation Complacency	MEDIUM	Nurses uncritically accepting AI outputs	Prominent disagreement badges; mandatory override reasons
5. Missing Data Exploitation	MEDIUM	Nurses skipping BP / vitals during rush	`UNKNOWN ≠ NORMAL` elevates uncertainty to trigger review
6. Regulatory Barriers	HIGH	Stringent SaMD compliance requirements	Early engagement with FDA / CDSCO; hybrid CDSS posture
7. Cybersecurity & HIPAA	HIGH	Hospital network breach or PHI leakage	Local in-browser execution; zero cloud transmission of PHI

Proactive Risk Governance: Safety risks are addressed at the architectural level through deterministic safety floors and local edge execution.

13. Scalability Architecture & Commercialization Models

All commercialization models outlined below represent **illustrative future business hypotheses** for prospective evaluation, not current commercial revenue.

Three Potential Commercialization Models

POTENTIAL COMMERCIALIZATION MODELS	
MODEL TYPE	STRUCTURE & OPERATIONAL FIT
1. Hospital SaaS License	Annual subscription per ED site, tiered by annual patient intake volume (e.g., <30k, 30k-70k, >70k visits).
2. Health System Enterprise	System-wide deployment covering multiple hospital sites with centralized analytics and cross-site telemetry.
3. EHR App Marketplace	Direct integration add-on within Epic App Orchard / Oracle Cerner Marketplace via SMART-on-FHIR standard.

Technical Scalability Advantages

- **Zero GPU Server Overhead:** In-browser TypeScript tree evaluation executes locally on triage workstations, eliminating expensive server-side inference infrastructure.
- **Configurable Hospital Profiles:** Built-in adaptation for Trauma Centers (10m wait caps), Community EDs (15m wait caps), and Rural Facilities (20m wait caps).
- **Modular Integration:** Compatible with existing hospital single sign-on (SSO), role-based access control, and audit pipelines.

14. Prototype Evidence & Hospital Integration Feasibility

Verified Synthetic Benchmark Metrics (2,250 Record Holdout Set)

 90.79% Synthetic Holdout Accuracy	 95.99% Critical Class Recall
 98.87% High + Critical Sensitivity	 0 Cases (0.0%) Critical → Low Misses

Exploratory External Data Analysis: Evaluated on a sample of 207 labeled ESI 1–4 records from the MIMIC-IV-ED database (`ml/data/triage.csv.gz`), the domain-robust feature extraction identified 88.89% (16/18) of Critical resuscitation cases (ESI Level 1) and 61.74% (71/115) of High+Critical cases zero-shot without fine-tuning. *(Note: This exploratory analysis demonstrates prototype transferability under missing data conditions, but does not constitute clinical validation.)*

Hospital EHR Integration Feasibility



- **Standard FHIR Resources:** Cleanly maps to standard FHIR `Patient` , `Observation` (vitals), `Condition` (complaints), `RiskAssessment` (triage), and `AuditEvent` (logs).
- **Zero Infrastructure Footprint:** Runs in modern web browsers without requiring local software installation or specialized client hardware.

15. Conclusion & Strategic Call to Action

Emergency department triage cannot remain a static, one-time snapshot while patients face extended waiting times in crowded waiting rooms.

PatientTriage.ai establishes a viable, transparent, and clinically responsible path forward. By subordinating machine learning to deterministic physiological safety rules and anchoring all decisions in licensed clinician authority, the platform demonstrates how artificial intelligence can expand clinical situational awareness without compromising safety.

THE CORE CALL TO ACTION

PatientTriage.ai

"See risk before it becomes a crisis."

- Transform static triage into continuous safety surveillance
- Enforce deterministic safety floors that ML cannot compromise
- Empower clinicians with explainable, human-in-the-loop intelligence

Strategic Next Step

Initiate formal retrospective clinical studies on governed multi-center EHR cohorts, followed by IRB-approved silent shadow trials in academic emergency departments.

PatientTriage.ai — Clinical Decision-Support Prototype — Synthetic Demonstration Data — Not Approved for Clinical Use